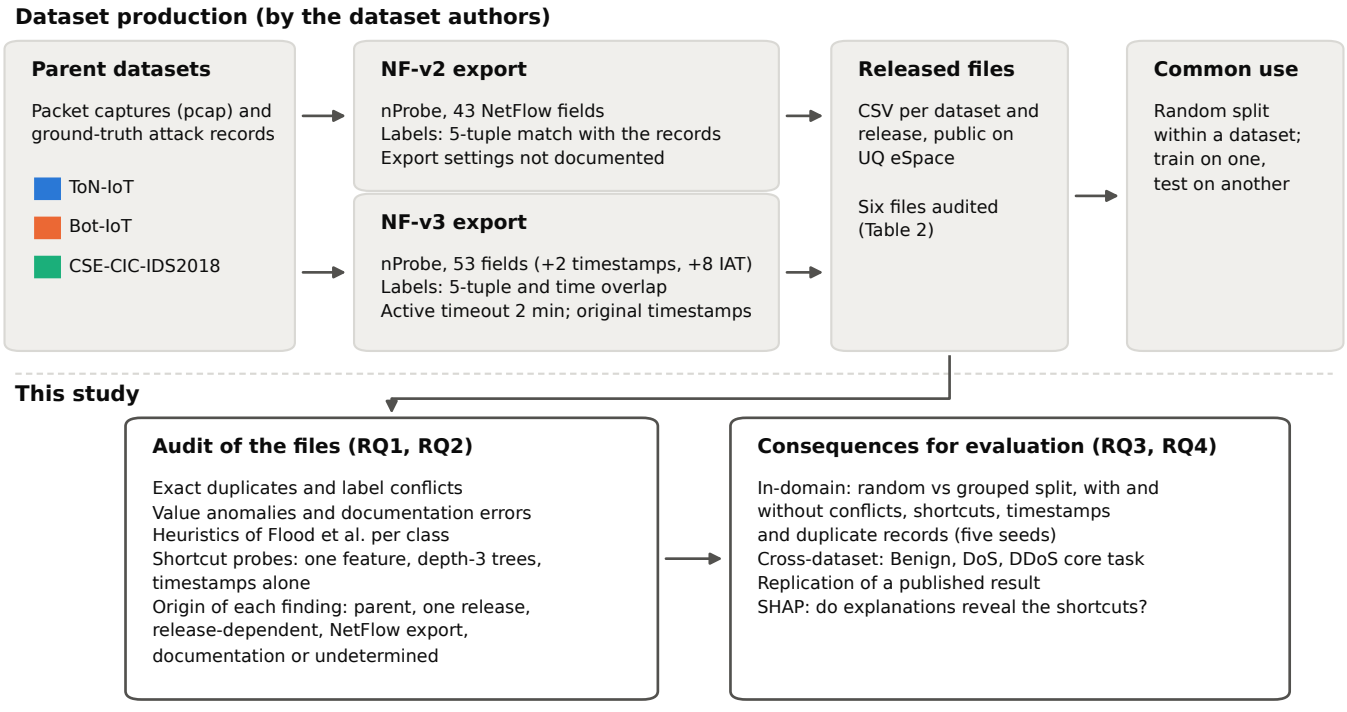

Dataset production (by the dataset authors)
Parent datasets
Packet captures (pcap) and ground-truth attack records
ToN-IoT
Bot-IoT
CSE-CIC-IDS2018
NF-v2 export
nProbe, 43 NetFlow fields
Labels: 5-tuple match with the records
Export settings not documented
NF-v3 export
nProbe, 53 fields (+2 timestamps, +8 IAT)
Labels: 5-tuple and time overlap
Active timeout 2 min; original timestamps
Released files
CSV per dataset and release, public on UQ eSpace
Six files audited (Table 2)
Common use
Random split within a dataset; train on one, test on another
This study
Audit of the files (RQ1, RQ2)
Exact duplicates and label conflicts
Value anomalies and documentation errors
Heuristics of Flood et al. per class
Shortcut probes: one feature, depth-3 trees, timestamps alone
Origin of each finding: parent, one release, release-dependent, NetFlow export, documentation or undetermined
Consequences for evaluation (RQ3, RQ4)
In-domain: random vs grouped split, with and without conflicts, shortcuts, timestamps and duplicate records (five seeds)
Cross-dataset: Benign, DoS, DDoS core task
Replication of a published result
SHAP: do explanations reveal the shortcuts?